Lloyd Institute of Engineering & Technology

Department of Computer Science & Engineering (AIML) | Session 2026-27

MINI PROJECT - WEEKLY PROGRESS LOG

Student Name	Dev Rana Kaila	**Roll Number**	2301531530023
Branch / Semester	B.Tech CSE (AIML) - Sem VII	**Section**	A
Project Guide	Dr. Devendra Gautam	**Designation**	Associate Professor, Dept. of CSE
Project Title	Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection		
GitHub	github.com/DevRanaKaila/DeepGuard-AI	**College**	LIET, Greater Noida

Weekly Activity Log (Guide Signatures Required)

Date Range	Phase / Milestone	Activities & Deliverables	Status	Guide Signature
06/09/2026 - 12/09/2026	**Topic Finalisation & Literature Study**	Finalised project topic on deepfake detection. Studied 8 research papers (Rossler, Frank, Qian, Afchar, Li, Tan, Selvaraju, Durall). Identified FaceForensics++ and Celeb-DF v2 as benchmark datasets. Set up Python environment: PyTorch 2.0+, timm, OpenCV, SciPy, Streamlit.	Completed	
13/09/2026 - 19/09/2026	**Synopsis Preparation**	Prepared formal MINI PROJECT SYNOPSIS in LIET-prescribed format. Defined 6 project objectives. Drafted Literature Survey, Work Plan, Hardware/Software requirements. Submitted draft to Dr. Devendra Gautam for review.	Completed	
20/09/2026 - 26/09/2026	**Synopsis Approval & Architecture Design**	Incorporated mentor feedback. Received synopsis approval. Designed dual-stream pipeline diagram, DFD Level-0 and Level-1. Initialized GitHub repository: github.com/DevRanaKaila/DeepGuard-AI. Initial commit with project structure.	Completed	
27/09/2026 - 03/10/2026	**Dataset Curation & Preprocessing Pipeline**	Set up FaceForensics++ dataset structure across raw, c23, c40 tiers. Implemented video frame extraction at 5 FPS using OpenCV. Integrated MTCNN for face detection with 15% boundary margin. Verified preprocessing pipeline on 50 sample videos. Created src/data/ module.	Completed	
04/10/2026 - 10/10/2026	**Spatial Stream Implementation**	Integrated EfficientNet-B4 backbone via timm (pretrained ImageNet). Implemented SpatialStream module in src/models/spatial_stream.py. Validated 1792-d feature vector output and gradient flow. Spatial-only baseline achieves 82% accuracy.	Completed	
11/10/2026 - 17/10/2026	**DCT Transform & Mid-Term Review**	Implemented 2D-DCT-II transform and Radial Azimuthal Spectrum A(r) in src/data/dct_transform.py. Demonstrated spatial pipeline to mentor. Prepared mid-term progress	Completed	

		report. Frequency stream CNN prototype initiated. Mid-term review signed off.		
18/10/2026 - 24/10/2026	**Frequency Stream & Fusion Module**	Implemented 4-stage Spectral-CNN (src/models/frequency_stream.py). Implemented attention-gated bilinear fusion (src/models/fusion.py). Complete DualStreamNet integration. Configured training engine with AdamW, label smoothing, and AMP.	Completed	
25/10/2026 - 31/10/2026	**Dashboard, Grad-CAM & Testing**	Develop Streamlit web dashboard (app.py). Implement Grad-CAM heatmap generator (src/utils/gradcam.py). End-to-end pipeline integration. Cross-dataset testing on Celeb-DF v2. Benchmark against XceptionNet and F3-Net baselines.	Planned	
01/11/2026 - 07/11/2026	**Report Writing & Code Finalization**	Draft complete 60+ page project report strictly adhering to AKTU formatting rules. Conduct ablation study (spatial-only vs. dual-stream). Run plagiarism check (Turnitin < 20%). Finalize GitHub repo with full documentation and README.	Planned	
08/11/2026 - 14/11/2026	**Final Submission & Viva Defense**	Spiral-bind project report. Prepare 15-slide viva presentation deck. Live demo of Streamlit dashboard with real and fake test videos. Final submission to Dr. Devendra Gautam. Panel question-and-answer defense.	Planned	

Note: Weeks marked Completed have been verified and signed off by the Project Guide. Remaining weeks will be evaluated upon milestone completion.

Student Signature: _____

Dev Rana Kaila | Roll No. 2301531530023

Guide Signature: _____

Dr. Devendra Gautam | Assoc. Professor, CSE